Uncertainty Quantification In Machine Learning Based Retrieval Of Soil Moisture From GNSS-R Observations

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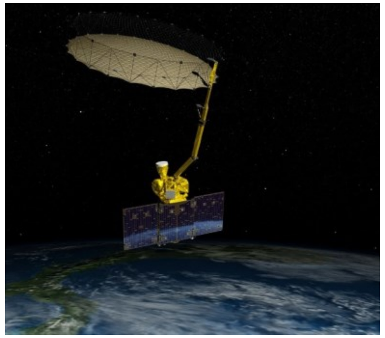
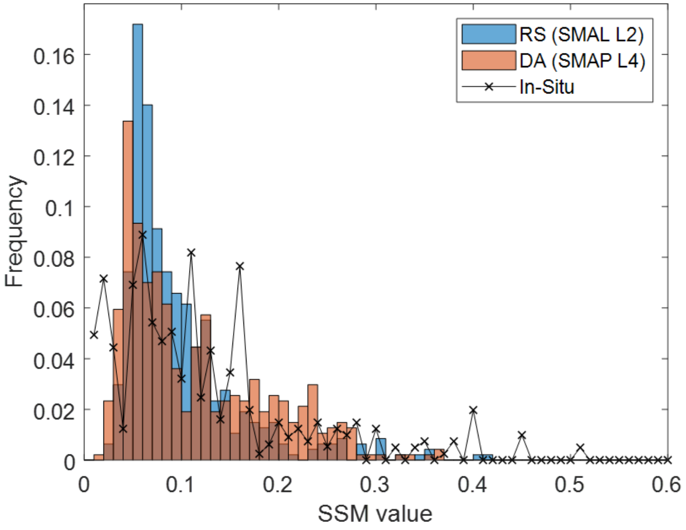
Remote Sensing Soil Moisture Retrieval



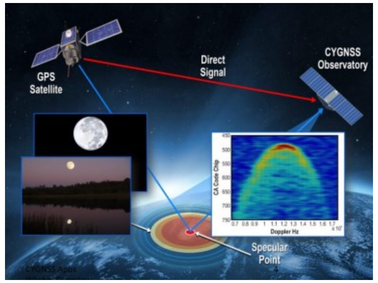
SMAP: global soil moisture every 2-3 days
L-band (1.2 GHz , 1.41 GHz)
- ➢ Radar (active) @ 3km
- ➢ Radiometer (passive) @ 36km



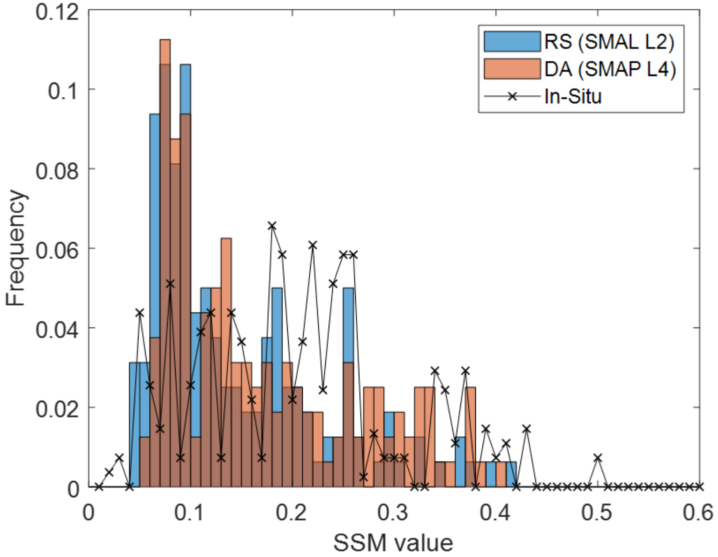
June 2017

ESA Sentinel 1 A/B: 6-12 days revisit
- C-band radar (5.405 GHz)

- ➢ SMAP/Sentinel 1 SM @ 3km



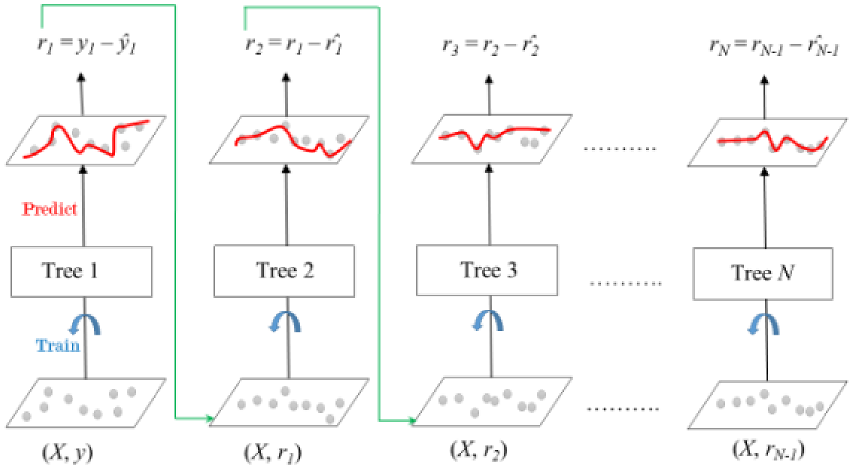
CYGNSS 8 satellites @ 35° orbit inclination
- Sea level wind speed in tropical cyclones
- Application in Surface Soil Moisture



June 2019

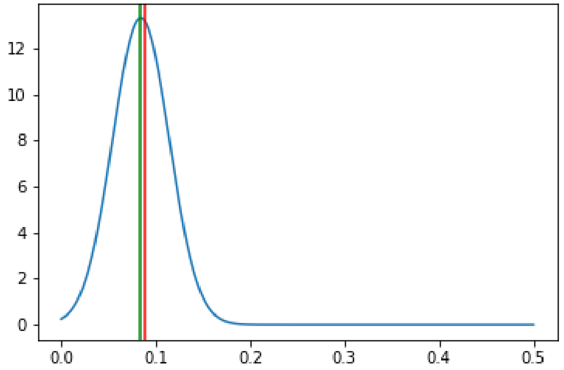
State-of-the-art:

- ➢Gradient boosting: Ensemble of weak prediction models, typically decision trees



- ➢XGBoost: regularized (L1 & L2) Gradient Boosting

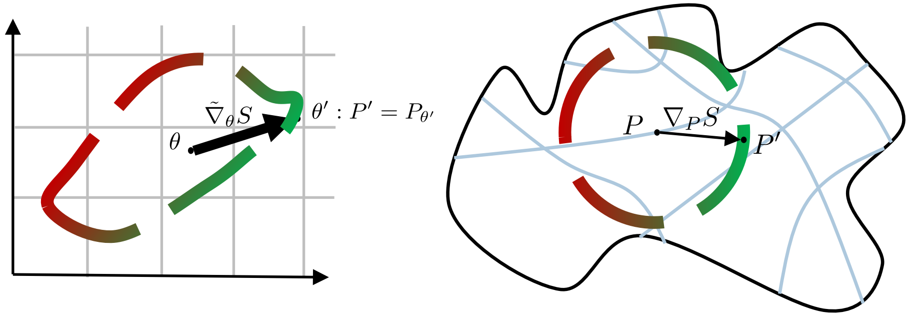
Question	ML answer
What is the Soil Moisture	0.02 cm^3/cm^3

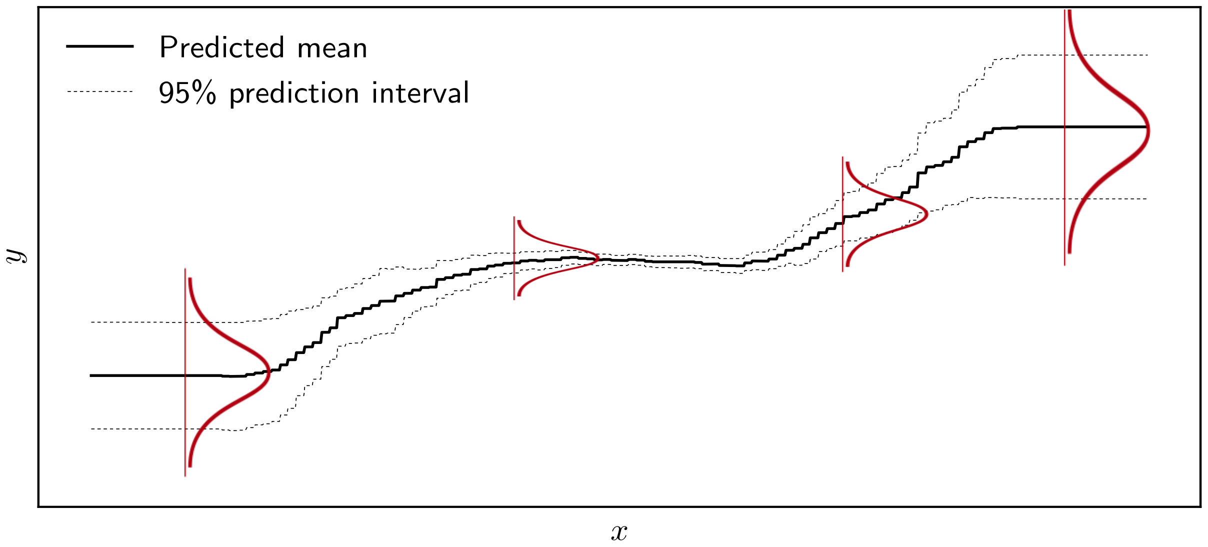


NGBoost

Natural gradient boosting for probabilistic prediction

$$\tilde{\nabla}_\theta S = I(\theta)^{-1} \nabla_\theta S$$





Point Prediction	Loss Function	$L(\hat{y}, y)$
Probabilistic Prediction	Scoring Rule	$S(\hat{P}, y)$

negative log-likelihood

$$S(\hat{P}, y) = -\log \hat{P}(y)$$

Duan, T., Anand, A., Ding, D. Y., Thai, K. K., Basu, S., Ng, A., & Schuler, A. ICML 2020

Experimental setup

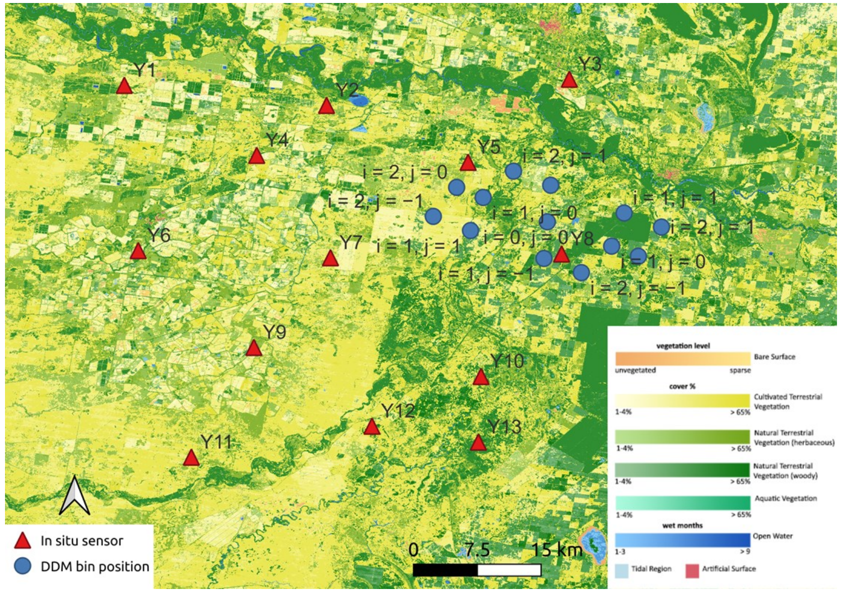
Remote sensing: CYGNSS DDM

In-situ: <5cm

Location: Yanco, Southeast Australia

Time: 2019-2020

Co-location: <5km, <6h

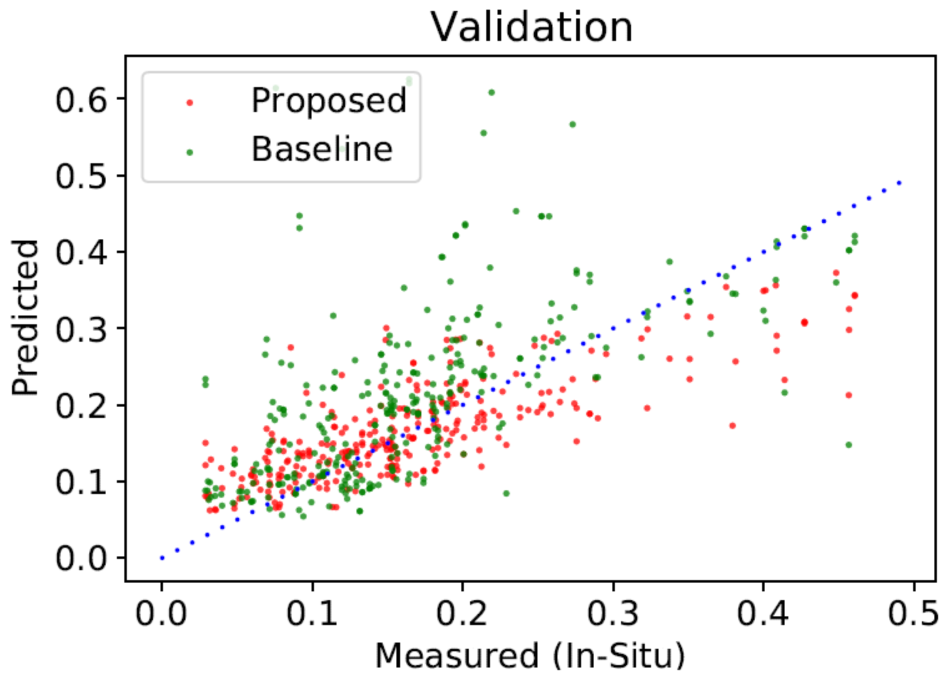


CYGNSS QUALITY FLAGS

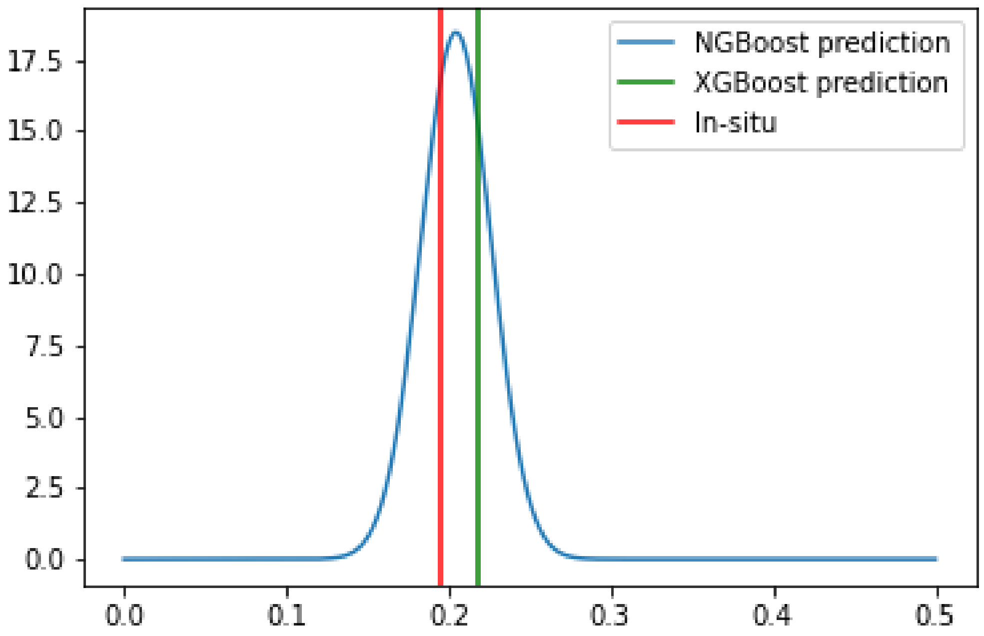
s_band_powered_up	small_sc_attitude_err
large_sc_attitude_err	black_body_ddm
ddmi_reconfigured	spacewire_crc_invalid
ddm_is_test_pattern	channel_idle
large_step_lna_temp	sp_non_existent_error

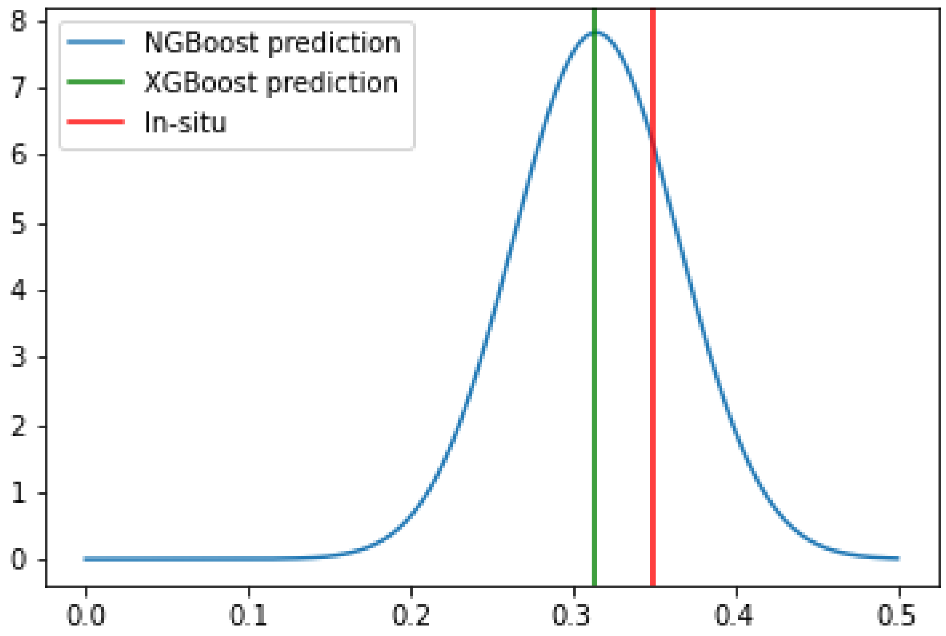
Retrieval *accuracy*

	RMSE $[m^3/m^3]$	ubRMSE $[m^3/m^3]$
SMAP (baseline)	0.103 (0.005)	0.151 (0.007)
CYGNSS (XGBoost)	0.055 (0.002)	0.059 (0.001)
CYGNSS (NGBoost)	0.058 (0.003)	0.060 (0.001)

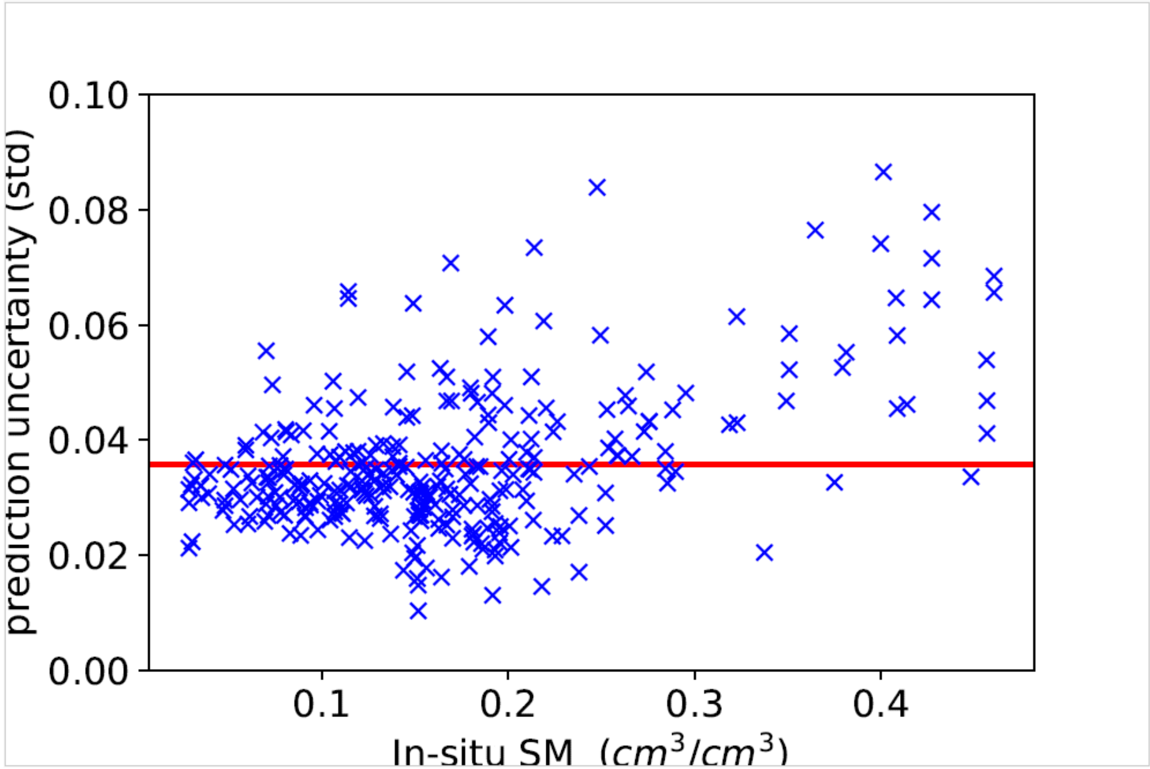


Challenge: quantifying performance

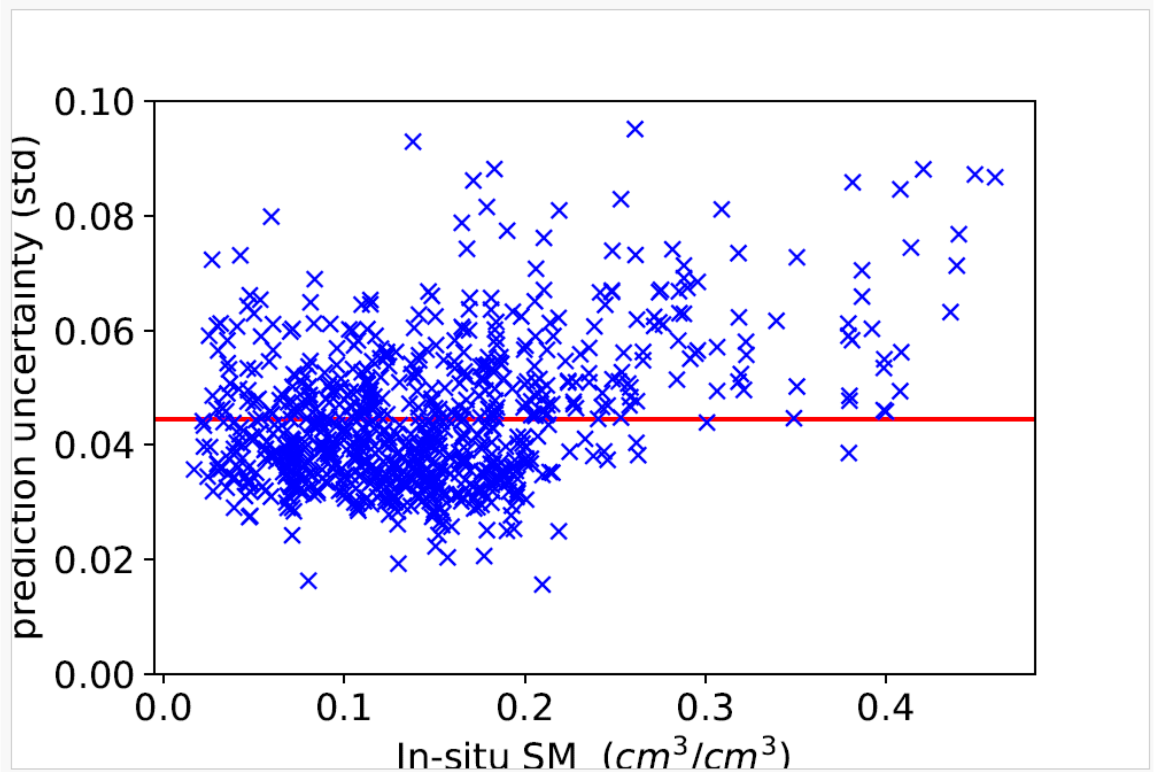




“Introducing” noise



SNR>10 dB

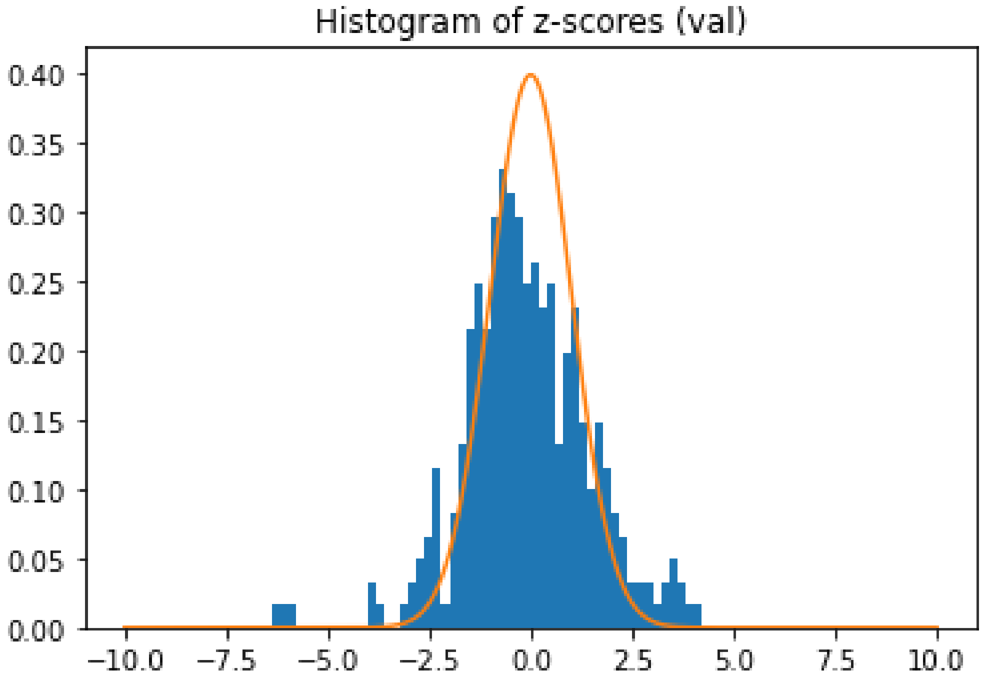


SNR>0 dB

Coverage probability

$$\mathrm{CP} = \frac{1}{n}\sum_{i=1}^{n} I\left(y_i \in \left[\hat{y}_i - z_{1-\frac{\alpha}{2}}\hat{\sigma}_i, \hat{y}_i + z_{1-\frac{\alpha}{2}}\hat{\sigma}_i\right]\right),$$

Significance level (α)	0.31	0.05	0.01
Gaussian	68.2	95.4	99.7
Training	63.7	89.6	94.2
Validation	63.1	89.6	93.8



Discussion

- Understand the impact of uncertainty
- Modeling physical processes
- Epistemic vs Aleatoric uncertainty
- Creating Analysis Ready Datasets

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